

Lecture 1 - Benefits of Feedback

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Feedback Control

Feedback control is a powerful concept in engineering, and systems in general, that enable effecting the dynamics of some system, or plant, in a *potentially* favorable way. We say potentially, because an improperly designed feedback control system can actually lead to a worse system response. The ways to navigate and prevent this will be covered later in the course. In simple terms, feedback control utilizes the output, or quantity that we want to control, to inform the control inputs used on the dynamical system of interest. Some examples of these are:

- Attitude tracking controller for a satellite (covered in the Feedback control of dynamic systems, Franklin, Powell, Emami-Naeini text)
- Temperature control (often utilize a "bang-bang" controller)
- Cruise control (simplified model covered in this lecture)

Some of the basic components of a feedback control structure are:

- Reference: Desired value of the quantity you want to control.
- Input: The commands to the actuators on the system.
- Plant: The dynamics of the system.
- Disturbances: External inputs to the plant that you do not have control over, such as bumps on a road or wind gusts.
- Output: The quantity or quantities you are measuring and using (in closed-loop control).

The typical feedback control structure for open-loop and closed-loop feedback is shown in Figure 1.1 below, with the formal definitions of open-loop and closed-loop control being discussed later in the lecture/course.

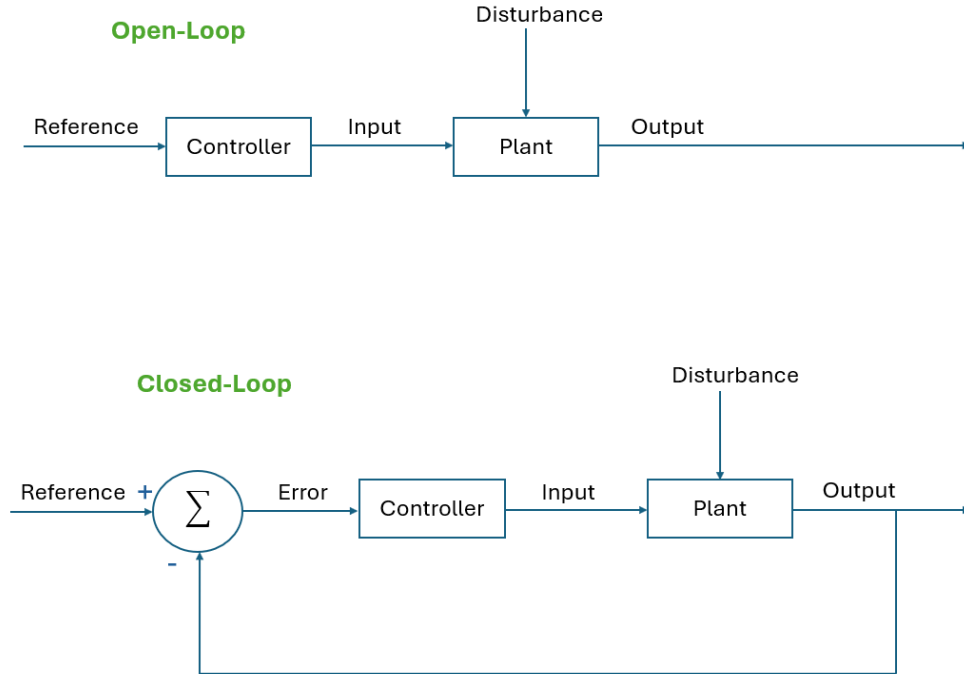


Figure 1.1: Block diagram structure for open-loop and closed-loop feedback control.

To provide some intuition on a simple feedback control structure and the differences between open-loop and closed-loop control, a motivating example is shown in Figures 1.2, 1.3, and 1.4.

Goal: Illustrate the benefits of feedback using a simple example.

Example: Cruise control accounting for slopes on the road.

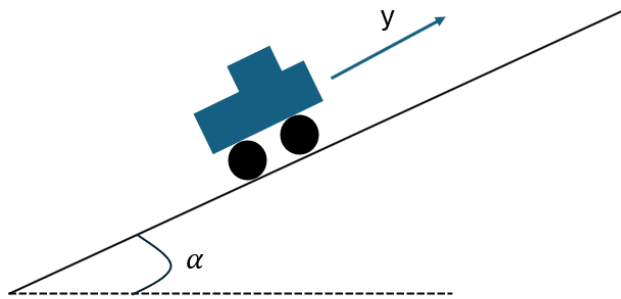


Figure 1.2: Cruise control problem setup.

Where we have:

- y : car speed (output)
- α : slope of road (disturbance)
- u : throttle angle (control input)

- r : reference speed (desired: 65 mph)

Goal: Choose u (our input) to set our speed y equal to our reference speed r .

Problem Formulation:

We need a mathematical model that relates y to r , considering u and α . Typically, this is done by deriving the equations of motion and taking the Laplace transform. However, for this simplified problem, we assume that the dynamics are simply unity, that is, the transfer function for the plant model is $G(s) = 1$.

- On a flat road, a 1° increase in the throttle angle (u) gives an increase in speed (y) of 5 mph
- On a slope, a 1° increase in slope angle (α) gives a decrease in speed (y) of 5 mph
- Ignore dynamic effects ($G(s)$, not shown, is set = 1)
- Want to stabilize y to equal r and reject the disturbances α

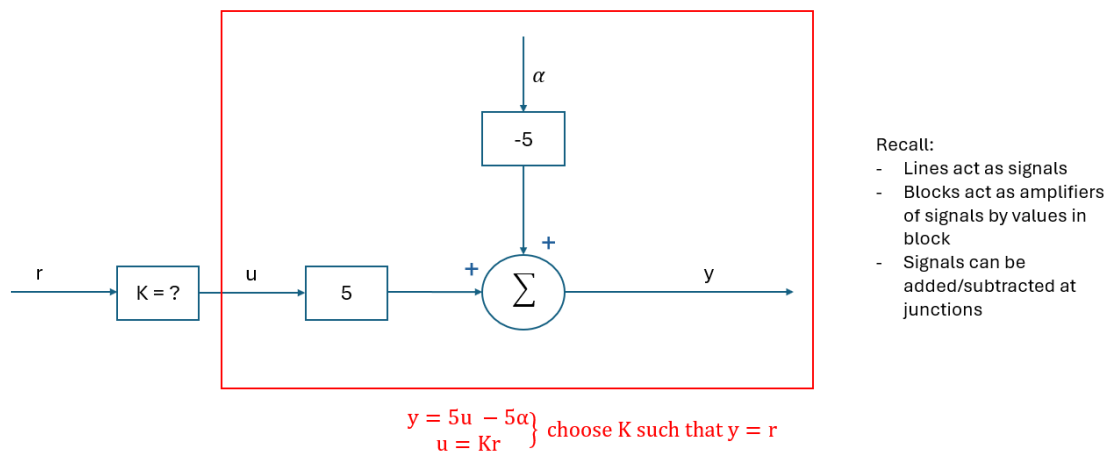


Figure 1.3: Block diagram of the cruise control problem.

1.1 Open-Loop Control Design

Open-loop control does not use any information about the current state of the system, it only uses the reference signal (will discuss feedforward aspects in future lectures). Hence why it is called "open-loop", the output is never fed back into the controller/system.

- We don't have a measurement of our speed y or the slope α
- But we drive mostly on flat roads, so we ignore α , or let $\alpha = 0$
 $\rightarrow y = 5u - 5 \cdot 0 = 5u$
- For our design requirement: $y = r$
 We design u s.t. $5u = r \rightarrow u = \frac{r}{5} \rightarrow K = \frac{1}{5}$
- $K = \frac{1}{5}$ is our open-loop controller

How well does our open-loop controller work?

- On flat ground, when $\alpha = 0 \rightarrow$
 $y = 5u - 5\alpha = 5 \cdot \frac{1}{5} \cdot r = r \rightarrow$ Great, exact reference tracking
- But on a slope, when $\alpha \neq 0 \rightarrow$
 $y = 5 \cdot r \cdot \frac{1}{5} - \alpha = r - 5\alpha \rightarrow$ Not so great
 Notice that the larger the slope, the larger the error e between y and r .

For this open-loop controller, we can look at:

- Tracking error, $e = y - r = 5\alpha$, where the error is proportional to the slope angle
- Normalized tracking error $= \frac{r-y}{r} = \frac{5\alpha}{r}$, showing that the error w.r.t our reference gets worse for large slopes (α) and low speeds (r)
- If $\alpha = 1^\circ$ and $r = 100$ mph:

$$\text{normalized error} = \frac{5 \cdot 1}{100} = \frac{5}{100} \rightarrow 5\%$$

- If $\alpha = 4^\circ$ and $r = 25$ mph:

$$\text{normalized error} = \frac{5 \cdot 4}{25} = \frac{20}{25} \rightarrow 80\%$$

Lesson:

- Open-loop controller is bad at rejecting disturbances (varying slopes)
- Open-loop controller performance above depends on $K = \frac{1}{5}$ perfectly canceling our plant gain of 5
 $\rightarrow$ if the plant gain changes (e.g. wear & tear), then the error gets worse

1.2 Closed-Loop Control Design

Closed-loop control utilizes a measurement of the output, or the quantity that you want to control, to inform the controller on what the input to the system should be. The block diagram for the closed-loop cruise control system is shown below.

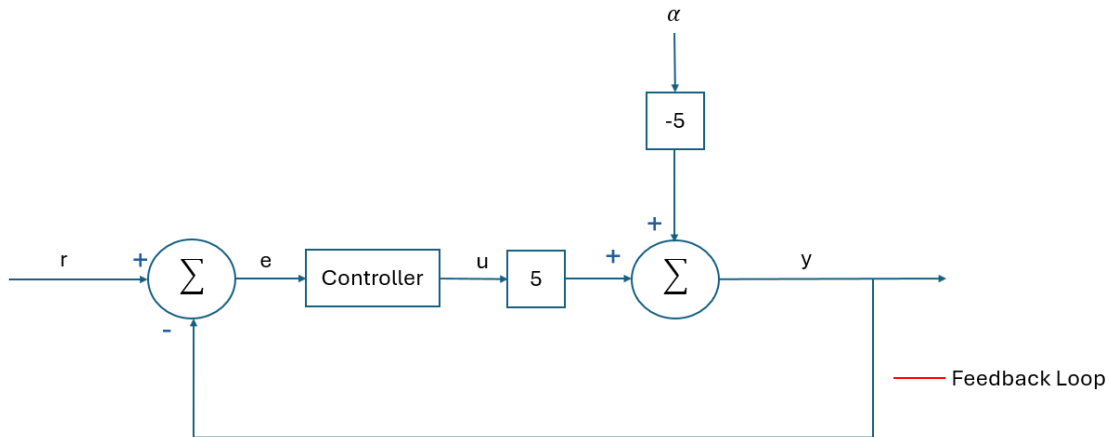


Figure 1.4: Closed-loop feedback block diagram for the cruise control example.

For the closed-loop feedback system, we define:

- Error, $e = r - y$
- The control u with control gain k , $u = ke$. This form of control is known as "proportional" control, as the control signal is proportional to the error.

Then, the closed-loop model can be found,

$$\begin{aligned} u &= ke = k(r - y), \\ y &= 5u - 5\alpha = 5k(r - y) - 5\alpha \end{aligned}$$

Rearranging for y in terms of r :

$$\begin{aligned} \rightarrow (1 + 5k)y &= 5kr - 5\alpha \\ \rightarrow y &= \frac{5k}{1 + 5k}r - \frac{5}{1 + 5k}\alpha \end{aligned} \tag{1.1}$$

The term multiplying r in (1.1) is the transfer function from the reference r to the output y (G_{ry}) and the term multiplying α in (1.1) is the transfer function from the disturbance α to the output y ($G_{\alpha y}$). These will be discussed in more detail in future lectures.

Now, how well can the output y track the reference r for the closed-loop system in (1.1)? We can take a look at the normalized tracking error,

$$\begin{aligned} \frac{r - y}{r} &= \left(1 - \frac{5k}{1 + 5k}\right) + \left(\frac{5}{1 + 5k}\right)\frac{1}{r}\alpha \\ &= \frac{1}{1 + 5k} + \frac{5\alpha}{r} \left(\frac{1}{1 + 5k}\right) \end{aligned}$$

Notice that the open-loop normalized tracking error ($\frac{5\alpha}{r}$) appears in the closed-loop normalized tracking error. Is this better than the open-loop normalized tracking error? To answer this, let's choose a value for the control gain k and look at the closed-loop normalized error.

Choose $k = 20 \rightarrow$ normalized error $\approx \frac{1}{100} + \frac{5\alpha}{r} \cdot \frac{1}{100}$.

The open-loop error ($\frac{5\alpha}{r}$) effect was decreased by $100\times$, however, the $\frac{1}{100}$ is a new added constant error bias. However, notice that k is in the denominator of both of these terms. **This means that the error decreases as k increases.**

- We will see that we cannot just increase k indefinitely because this leads to instability, as well as physical constraints such as leading to too large of a control input for the actuators.
- Later we will use proportional-integral (PI) control to make the bias error = zero.

Lesson:

- Feedback control makes performance less sensitive to disturbances.
- We don't depend on having a perfect model of the plant to "invert" for the control law.